

Lab 2 : MultiCycle Performance

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0.1 Matrix Multiplication Performance

0.1.1 Procedure

1. For each matrix test case (3x3, 10x10, 16x16, 50x50) ...
2. Compile the matrix multiplication program in the OTTER-memory mapped version of RARS. The RARS program should compile but not run any instructions of assembly. The machine code file needs to be saved as a .mem
3. Load the mem file into the multicycle OTTER in Vivado and simulate.
4. Capture the waveforms for the matrix multiplication to prove the correct execution.
5. Provide CPU execution time , Power, and Area utilization.

0.2 Testall Performance

0.2.1 Procedure

- 1. Load the testall mem file onto the multicycle otter. It should be noted that Diego’s testall program worked well such that the Basys3 board also gave LED and sseg indicators that the system was working.
- 2. Capture the same waveforms, cpu execution time, power, and area utilization.

0.2.2 Results

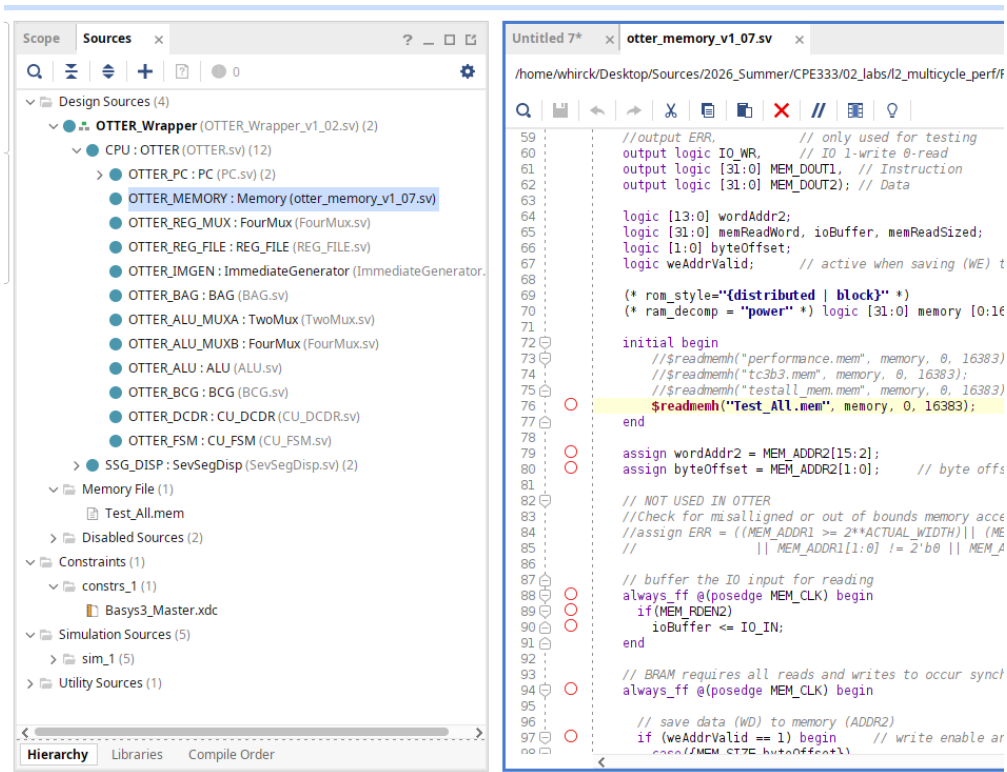


Figure 1: Uploading the testall memory

Timing

Clock Summary			
Name	Waveform	Period (ns)	Frequency (MHz)
sys_clk_pin	{0.000 5.000}	10.000	100.000

Figure 2: Clock Speed

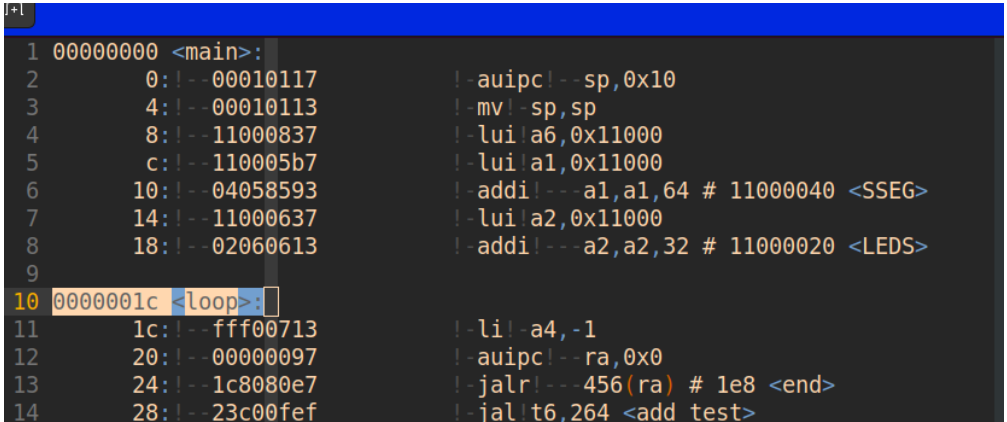


Figure 3: Target Instruction Address for Timing Intervals in Test_All_Debug.txt

The loop label was used for measurement because the testall program returns to this after running all tests.

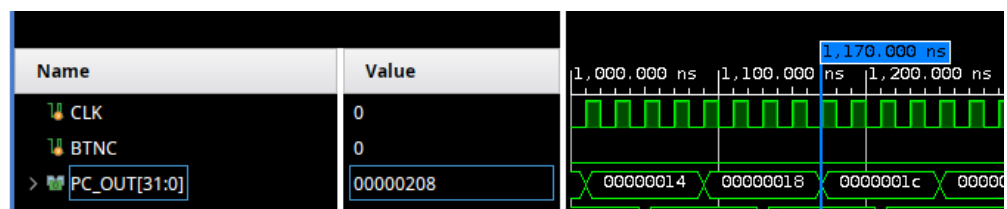


Figure 4: Measured Start Timestamp

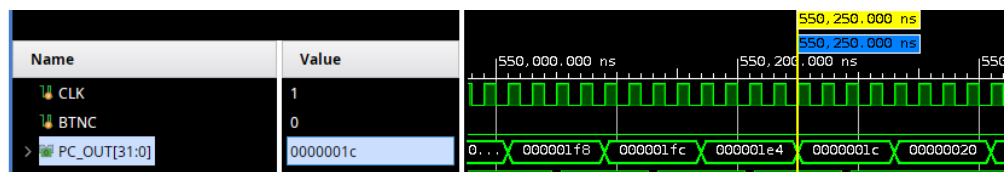


Figure 5: Measured End Timestamp

$$\begin{aligned}
 t_{\text{execution}} &= t_{\text{end}} - t_{\text{start}} \\
 &= 550,250\text{ns} - 1,170\text{ns} \\
 &= 549,080\text{ns}
 \end{aligned}$$

$$\begin{aligned}
 n_{\text{CLK_Cycles}} &= \frac{t_{\text{execution}}}{T_{\text{CLK}}} \\
 &= \frac{549080\text{ns}}{10\text{ns}} \\
 &= 54,908
 \end{aligned}$$

Note that the number above is not the number of instructions executed.

Other Metrics

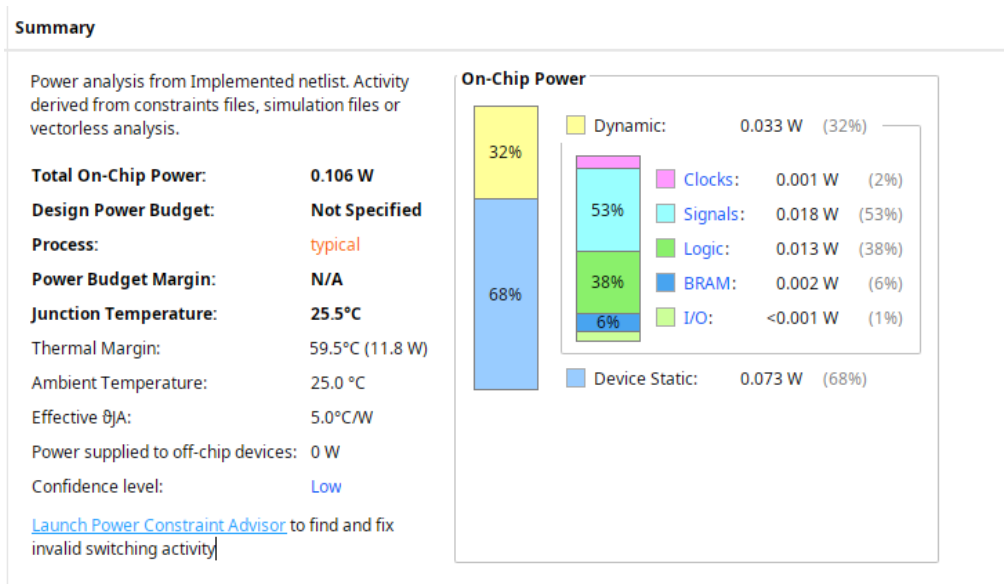


Figure 6: Power Summary of Testall on Basys3

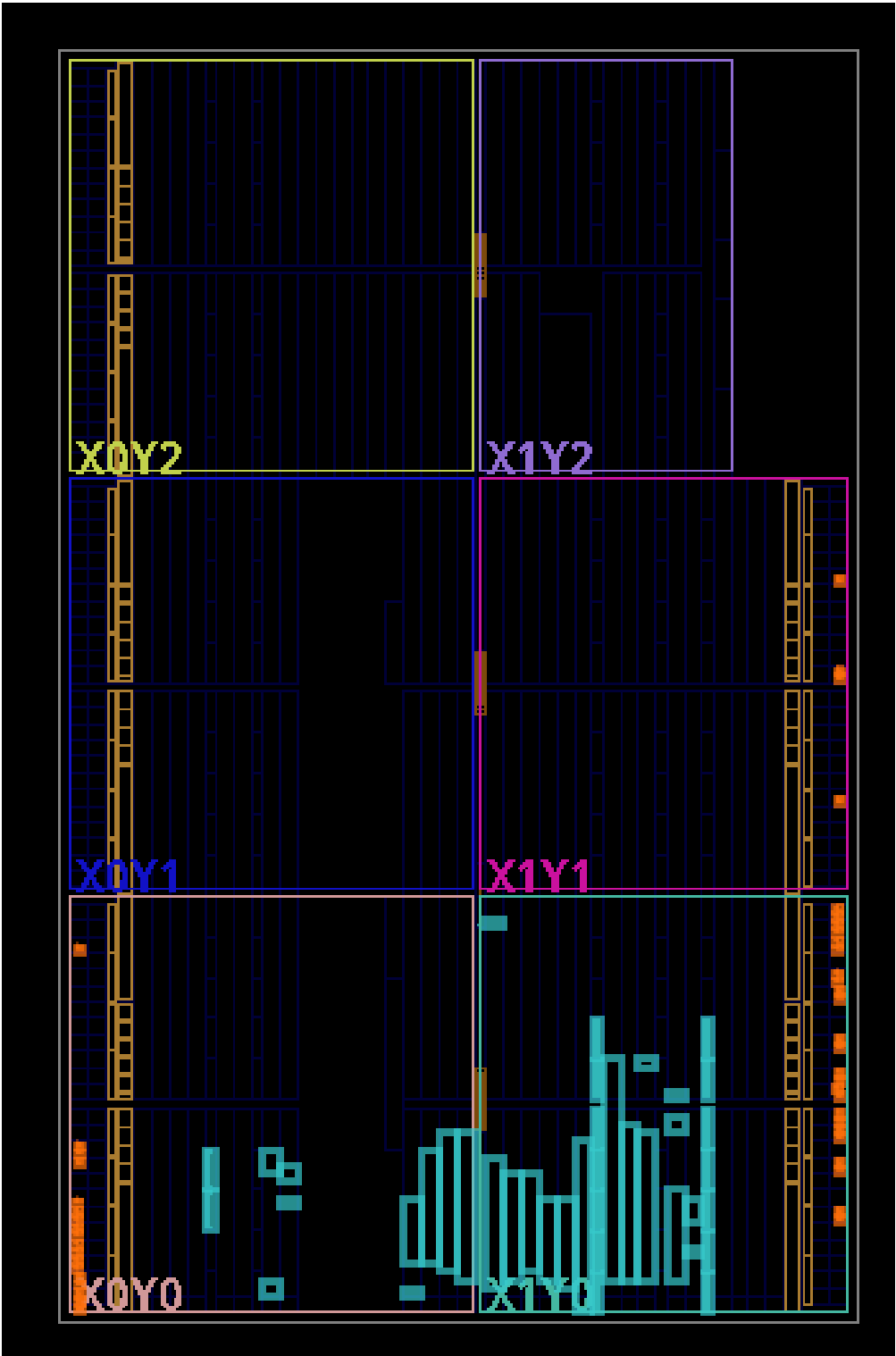


Figure 7: Visual Utilization Map of Testall

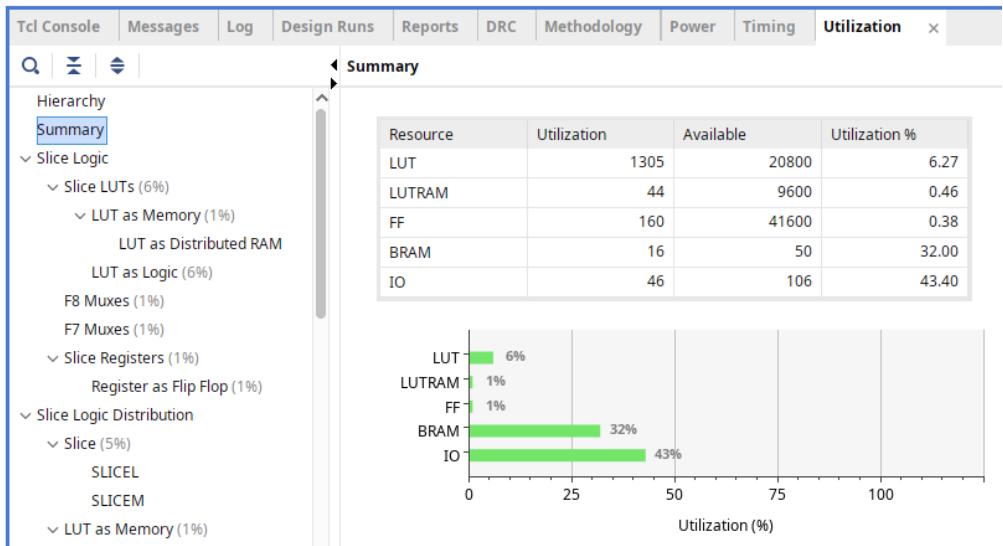


Figure 8: Utilization Summary of Testall

0.3 Notes

0.3.1 RARS

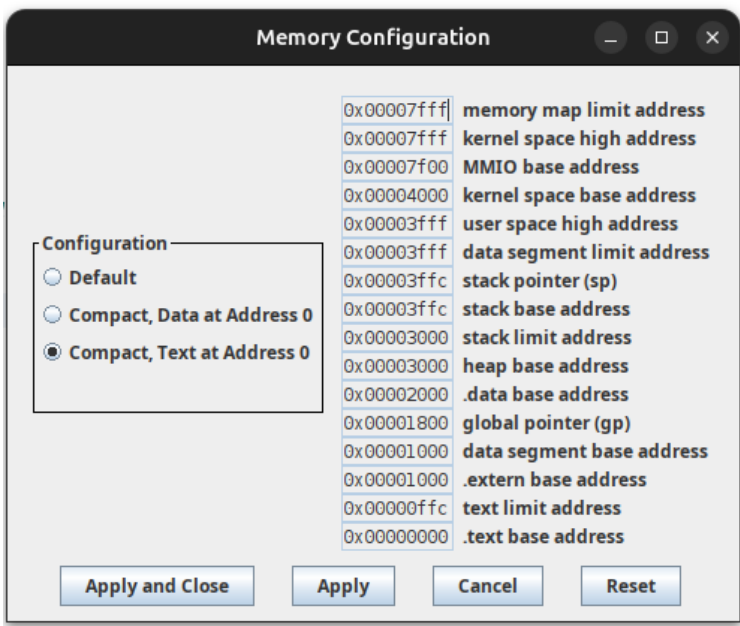


Figure 9: RARS Memory Config Settings

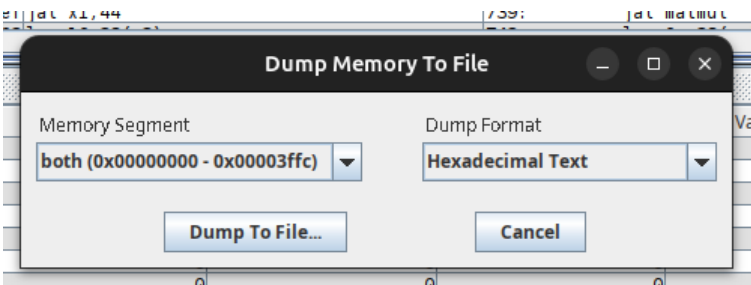


Figure 10: RARS Memory Dump Config